Understanding Conditions C3, C5, and C6 for Proposition 2 Aggregation Equivalence

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Recap: The Setup

The firm-level regression is

$$\mathbf{1}(\text{BNDES}_{fmt} > 0) = \sum_{\ell} \lambda_\ell \, \text{FA}^{\ell}_{fmt} + \gamma_f + \alpha_{mt} + u_{fmt}, \tag{1}$$

where f indexes firms, m municipalities, t years, and ℓ alignment tiers. The fixed effects are γ_f (firm) and α_{mt} (municipality×year).

Each firm belongs to a **cell** $c = (j, m, e)$, where j is the sector, m is the municipality, and e is the election cycle (support regime). The pre-election firm set in cell c is $\mathcal{F}^{\text{pre}}_c$ with $N_c = |\mathcal{F}^{\text{pre}}_c|$ firms.

Averaging equation (1) over firms in cell c gives the **cell-level equation**:

$$H^{\text{pre}}_{jmt} = \sum_{\ell} \lambda_\ell \, \overline{\text{FA}}^{\ell,\text{pre}}_{jmt} + \bar{\gamma}_c + \alpha_{mt} + \epsilon_{jmt}, \tag{2}$$

where $H^{\text{pre}}_{jmt} = N_c^{-1} \sum_{f \in \mathcal{F}^{\text{pre}}_c} \mathbf{1}(\text{BNDES}_{fmt} > 0)$, $\overline{\text{FA}}^{\ell,\text{pre}}_{jmt} = N_c^{-1} \sum_{f \in \mathcal{F}^{\text{pre}}_c} \text{FA}^{\ell}_{fmt}$, $\bar{\gamma}_c = N_c^{-1} \sum_{f \in \mathcal{F}^{\text{pre}}_c} \gamma_f$, and $\epsilon_{jmt} = N_c^{-1} \sum_{f \in \mathcal{F}^{\text{pre}}_c} u_{fmt}$.

Proposition 2 claims that running OLS on (1) (unweighted) and running N_c-weighted OLS on (2) produce the **same** $\hat{\lambda}$. This requires conditions C1–C6. Below we explain C3, C5, and C6 in detail.

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1 Condition C3: Firm Immobility

Condition 1 (C3 — Firm Immobility)**.** Each firm f belongs to exactly one cell (j, m) across all years in the sample. Formally:

$$\forall f: \quad \big|\{(j,m) : \exists\, t \text{ with } (f,j,m,t) \in \text{sample}\}\big| = 1.$$

1.1 Intuitive Explanation

Imagine you sort kids into classrooms based on their neighborhood and favorite subject. Each kid has a "personal talent score" (γ_f) that stays with them forever.

If every kid stays in the same classroom for the whole school, you can compute each classroom's "average talent" and use it as a single classroom label. No information leaks between classrooms.

But suppose Alice moves from Classroom A to Classroom B halfway through the year. Alice's talent score is *shared* between both classrooms. The per-kid regression knows this: it tracks Alice across both rooms and uses data from *both* rooms to estimate her talent. But when you compute classroom averages, each classroom computes its own average independently—Classroom A doesn't know that Alice also contributed to Classroom B's average. The per-kid regression and the classroom-average regression "see" different patterns, so they give different answers.

1.2 Mathematical Explanation

In the firm-level regression, γ_f is a single parameter estimated using *all* observations of firm f, regardless of which cells they belong to. The Frisch–Waugh–Lovell (FWL) projector M_{firm} absorbs $\{\gamma_f, \alpha_{mt}\}$ by projecting onto the orthogonal complement of the column space of the fixed-effect design matrix $[D_f \mid D_{mt}]$.

When C3 is violated. Suppose firm f operates in two cells, $c_1 = (j_1, m_1)$ and $c_2 = (j_2, m_2)$. Then γ_f enters the normal equations for *both* cells, creating a **cross-cell link**. When estimating $\hat{\gamma}_f$, OLS uses firm f's observations from both c_1 and c_2. This means the projected residual $\tilde{X}_{fmt}$ for an observation in c_1 depends on data from c_2.

After cell-averaging, the cell FEs are:

$$\bar{\gamma}_{c_1} = \frac{1}{N_{c_1}} \sum_{f' \in \mathcal{F}^{\text{pre}}_{c_1}} \gamma_{f'}, \qquad \bar{\gamma}_{c_2} = \frac{1}{N_{c_2}} \sum_{f' \in \mathcal{F}^{\text{pre}}_{c_2}} \gamma_{f'}.$$

Both contain γ_f (since $f \in \mathcal{F}^{\text{pre}}_{c_1} \cap \mathcal{F}^{\text{pre}}_{c_2}$), so they are *not independent*. However, the cell-level FWL projector M_{cell} treats $\bar{\gamma}_{c_1}$ and $\bar{\gamma}_{c_2}$ as **separate, independent parameters**: each cell's FE indicator column D_c is disjoint from every other cell's. It does not "know" that $\bar{\gamma}_{c_1}$ and $\bar{\gamma}_{c_2}$ share a component γ_f / N_{c_1} and γ_f / N_{c_2}, respectively.

As a result:

$$M_{\text{firm}} \neq \text{“cell-averaged version of } M_{\text{cell}}\text{”},$$

which means the projected cell means differ between the two approaches:

$$\frac{1}{N_c} \sum_{f \in c} \widetilde{X}^{\text{firm}}_{fmt} \;\neq\; \widetilde{\bar{X}}^{\text{cell}}_{ct},$$

and the normal equations no longer coincide even with N_c-weighting.

When C3 holds. If every firm belongs to exactly one cell, then D_f is *nested within* D_c: all of firm f's observations lie in a single cell. The firm FE becomes a within-cell parameter. Cross-cell links vanish, and the between-cell components of the two projections agree.

Example 1. In the data, about 20% of firms (accounting for 36% of observations) operate in more than one (j, m) cell. These multi-cell firms create exactly the cross-cell links described above.

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2 Condition C5: Fixed Cell Composition Within Regime

Condition 2 (C5 — Fixed Cell Composition Within Regime)**.** Within each support regime (election cycle) e, the set of firms in each cell is constant across all years t of that regime:

$$\forall\, c = (j, m, e),\ \forall\, t, t' \in \text{years}(e): \quad \mathcal{F}^{\text{pre}}_c(t) = \mathcal{F}^{\text{pre}}_c(t').$$

Equivalently, within each regime, every firm is present in **all** years of the regime (a balanced panel within each regime).

2.1 Intuitive Explanation

Imagine each classroom has a team roster that is supposed to be fixed for the whole school year (one election cycle). The classroom gets *one name tag*—the cell FE $\bar{\gamma}_c$—for the entire year.

But what if some kids join in March and others drop out in October? The classroom's "average talent" in September is different from its "average talent" in February, because the team changed. Yet we only have *one* name tag trying to represent the whole year. It captures a sort of "average of the averages," but it cannot track the month-to-month shifts.

The per-kid regression has no such problem: each kid has their own personal tag that stays constant regardless of what other kids do. Kids entering or leaving the classroom doesn't confuse anything.

So if the team roster shifts within a year, the one-tag-per-classroom approach cannot keep up, and the answers diverge.

2.2 Mathematical Explanation

From the averaging identity, the cell-level "firm FE" at time t is:

$$\bar{\gamma}_c(t) = \frac{1}{|\mathcal{F}_c^{\text{pre}}(t)|} \sum_{f \in \mathcal{F}_c^{\text{pre}}(t)} \gamma_f,$$

where $\mathcal{F}_c^{\text{pre}}(t)$ is the set of firms present in cell c at time t.

When C5 holds. $\mathcal{F}_c^{\text{pre}}(t) = \mathcal{F}_c^{\text{pre}}$ is constant within regime e, so $\bar{\gamma}_c(t) = \bar{\gamma}_c$ is a time-invariant constant (within the regime). A single FE per (j, m, e) triple absorbs it perfectly.

When C5 is violated. If firms enter or exit within the regime, $\bar{\gamma}_c(t)$ varies by year. The aggregated regression absorbs a single parameter δ_c per (j, m, e) cell—effectively the OLS "best constant approximation" to $\bar{\gamma}_c(t)$. The residual

$$\eta_{ct} \equiv \bar{\gamma}_c(t) - \delta_c$$

is time-varying and ends up in the error term ϵ_{ct}. If η_{ct} correlates with $\overline{\text{FA}}_{jmt}^{\ell,\text{pre}}$ (which it may, since both depend on *which firms are present* at time t), the cell-level OLS estimate $\hat{\lambda}^{\text{cell}}$ is biased *relative to* $\hat{\lambda}^{\text{firm}}$.

Example 2. Suppose cell c has firms $\{A, B\}$ in year 1 and $\{A, B, C\}$ in year 2 (firm C enters in year 2). Then:

$$\bar{\gamma}_c(1) = \tfrac{1}{2}(\gamma_A + \gamma_B),$$
$$\bar{\gamma}_c(2) = \tfrac{1}{3}(\gamma_A + \gamma_B + \gamma_C).$$

The cell FE δ_c can absorb at most one of these, not both. The leftover η_{c1} or η_{c2} contaminates the error term.

Meanwhile, $\overline{\text{FA}}_c(t)$ also shifts: in year 2, firm C's party exposure enters the cell mean. If firm C happens to be politically connected (high FA) *and* has high γ_C (unobservably productive), then $\eta_{c2} > 0$ and $\overline{\text{FA}}_c(2)$ is inflated, creating a spurious positive correlation.

In the firm-level regression, each γ_f is absorbed individually, so firm C's entry creates no bias.

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3 Condition C6: No Within-Cell Regressor Heterogeneity

Condition 3 (C6 — No Within-Cell Regressor Heterogeneity After FE Projection)**.** After absorbing fixed effects, the projected regressors are constant within each cell-year:

$$\tilde{X}_{fmt} = \bar{\tilde{X}}_{ct} \quad \forall\, f \in \mathcal{F}_c^{\text{pre}},\ \forall\, t,$$

where tildes denote FE-projected residuals and $\bar{\tilde{X}}_{ct} = N_c^{-1} \sum_{f \in c} \tilde{X}_{fmt}$.

Equivalently: *within-cell, between-firm variation in the projected* FA^{ℓ} *regressors is zero.*

3.1 Intuitive Explanation

Imagine two kids in the same classroom: Alice studies 5 hours and scores 90; Bob studies 1 hour and scores 60. The classroom average is "3 hours studied, 75 points scored."

The per-kid regression sees the *within-classroom* pattern: the kid who studied more scored more. It uses this Alice-versus-Bob contrast to learn that studying helps.

The classroom-average regression only sees *one data point* for this classroom: (3 hours, 75 points). It cannot look *inside* the classroom. The within-classroom variation—the Alice-vs-Bob difference—is invisible to it.

If both kids studied the same amount (say, both 3 hours, but Alice scored 90 and Bob scored 60 for other reasons), then the classroom average captures *all* the study-hour information perfectly—there is nothing hidden inside. Only then do the two approaches give the same answer.

This is why the synthetic test (which forces all "kids" in the same classroom to have the same "study hours") works perfectly at 10^{-15} precision, but real data (where firms in the same sector-municipality cell have different party exposures) never will.

3.2 Mathematical Explanation

After absorbing fixed effects via FWL, the firm-level OLS normal equation is:

$$\sum_c \sum_{f\in c} \tilde{X}'_{fmt}\big(\tilde{Y}_{fmt} - \tilde{X}_{fmt}\,\lambda\big) = 0. \tag{3}$$

Decompose each observation into its cell-year mean and within-cell deviation:

$$\tilde{X}_{fmt} = \underbrace{\bar{\tilde{X}}_{ct}}_{\text{between-cell}} + \underbrace{(\tilde{X}_{fmt} - \bar{\tilde{X}}_{ct})}_{\text{within-cell}}, \qquad \tilde{Y}_{fmt} = \bar{\tilde{Y}}_{ct} + (\tilde{Y}_{fmt} - \bar{\tilde{Y}}_{ct}).$$

Substituting into (3) and expanding (using the fact that within-cell deviations sum to zero), the normal equation splits into two orthogonal components:

$$\underbrace{\sum_c N_c\,\bar{\tilde{X}}'_{ct}\big(\bar{\tilde{Y}}_{ct} - \bar{\tilde{X}}_{ct}\,\lambda\big)}_{B(\lambda)\text{ — between-cell}} + \underbrace{\sum_c \sum_{f\in c}(\tilde{X}_{fmt} - \bar{\tilde{X}}_{ct})'\big((\tilde{Y}_{fmt} - \bar{\tilde{Y}}_{ct}) - (\tilde{X}_{fmt} - \bar{\tilde{X}}_{ct})\,\lambda\big)}_{W(\lambda)\text{ — within-cell}} = 0. \tag{4}$$

The N_c-weighted cell-level OLS normal equation is **only the between-cell part**:

$$B(\lambda) = \sum_c N_c\,\bar{\tilde{X}}'_{ct}\big(\bar{\tilde{Y}}_{ct} - \bar{\tilde{X}}_{ct}\,\lambda\big) = 0. \tag{5}$$

For both estimators to yield the same $\hat{\lambda}$, we need

$$B(\hat{\lambda}) + W(\hat{\lambda}) = 0 \quad \text{and} \quad B(\hat{\lambda}) = 0 \quad \Longrightarrow \quad W(\hat{\lambda}) = 0.$$

When does $W(\hat{\lambda}) = 0$? Define the within-cell cross-product matrices:

$$W_{XX} = \sum_c \sum_{f\in c}(\tilde{X}_{fmt} - \bar{\tilde{X}}_{ct})(\tilde{X}_{fmt} - \bar{\tilde{X}}_{ct})', \tag{6}$$

$$W_{XY} = \sum_c \sum_{f\in c}(\tilde{X}_{fmt} - \bar{\tilde{X}}_{ct})(\tilde{Y}_{fmt} - \bar{\tilde{Y}}_{ct}). \tag{7}$$

The within-cell normal equation $W(\lambda) = 0$ can be written as $W_{XY} - W_{XX}\,\lambda = 0$. This vanishes for all λ if and only if $W_{XX} = 0$ and $W_{XY} = 0$. And $W_{XX} = 0$ requires:

$$\tilde{X}_{fmt} = \bar{\tilde{X}}_{ct} \quad \forall\, f \in c,\ \forall\, t.$$

That is: after FE projection, every firm in the same cell-year must have the **same** projected FA^ℓ value. In other words, there must be *zero within-cell variation* in the projected regressors.

Why this fails on real data. The FA^ℓ_{fmt} instrument is the product of a firm-specific baseline party exposure ω_{fp} and a municipality-level alignment indicator Align^ℓ_{mpt}:

$$\text{FA}^\ell_{fmt} = \sum_p \omega_{fp,0}\,\text{Align}^\ell_{mpt}.$$

The alignment term Align^ℓ_{mpt} is common to all firms in the same (m,t), but the baseline exposure $\omega_{fp,0}$ varies across firms. Firms in the same cell (j,m,t) have different owners affiliated with different parties, so $\omega_{fp,0}$ differs across f within the cell.

After the firm FE projection (which removes each firm's time-mean), the projected regressor is:

$$\tilde{\text{FA}}^\ell_{fmt} = \text{FA}^\ell_{fmt} - \bar{\text{FA}}^\ell_{f.} - \bar{\text{FA}}^\ell_{.mt} + \bar{\text{FA}}^\ell_{..},$$

where $\bar{\text{FA}}^\ell_{f.}$ is firm f's time-mean. Since $\omega_{fp,0}$ varies across firms, $\bar{\text{FA}}^\ell_{f.}$ also varies across firms in the same cell, so $\tilde{\text{FA}}^\ell_{fmt}$ differs across firms within the cell. Therefore $W_{XX} \neq 0$.

Magnitude of the gap. The firm-level $\hat{\lambda}^{\text{firm}}$ solves $B(\lambda) + W(\lambda) = 0$, while the cell-level $\hat{\lambda}^{\text{cell}}$ solves $B(\lambda) = 0$. Using standard between/within decomposition algebra:

$$\hat{\lambda}^{\text{firm}} = (B_{XX} + W_{XX})^{-1}(B_{XY} + W_{XY}), \tag{8}$$

$$\hat{\lambda}^{\text{cell}} = B_{XX}^{-1}\, B_{XY}, \tag{9}$$

where $B_{XX} = \sum_c N_c\, \tilde{\tilde{X}}_{ct}\tilde{\tilde{X}}'_{ct}$ and $B_{XY} = \sum_c N_c\, \tilde{\tilde{X}}_{ct}\tilde{\tilde{Y}}_{ct}$.

The gap is:

$$\hat{\lambda}^{\text{firm}} - \hat{\lambda}^{\text{cell}} = (B_{XX} + W_{XX})^{-1}(B_{XY} + W_{XY}) - B_{XX}^{-1}B_{XY}.$$

This is zero if and only if $W_{XX} = 0$ and $W_{XY} = 0$. On real data, $W_{XX}/(B_{XX} + W_{XX})$ measures the *share of total variation that is within-cell*. The larger this share, the larger the potential gap.

Remark 1 (Why synthetic data achieves machine precision)**.** *In the synthetic benchmark,* FA^{ℓ}_{fmt} *is constructed to be constant within each cell-year:* $\mathrm{FA}^{\ell}_{fmt} = \mathrm{FA}^{\ell}_{ct}$ *for all* $f \in c$*. This forces* $W_{XX} = 0$ *and* $W_{XY} = 0$*, so* $\hat{\lambda}^{\text{firm}} = \hat{\lambda}^{\text{cell}}$ *at machine precision (*$\sim 10^{-15}$*).*

Remark 2 (C6 is the irreducible gap)**.** *Conditions C1–C5 can all be enforced on real data (by reweighting, restricting to single-cell firms, balancing the panel, using the correct FE, and disabling singleton removal). Condition C6* ***cannot*** *be enforced on real data without modifying the instruments themselves, because within-cell heterogeneity in* $\omega_{fp,0}$ *is an intrinsic feature of the data. The residual gap after enforcing C1–C5 (the "Silver tier" gap) is entirely attributable to C6.*

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4 Summary

#	Condition	What breaks without it	Enforceable on real data?
C3	Firm immobility (one cell per firm)	Firm FE creates cross-cell links that the cell-level projector ignores	Yes — restrict to single-cell firms (drops ∼20% of firms)
C5	Fixed composition within regime	Cell FE $\bar{\gamma}_c(t)$ becomes time-varying; single cell FE δ_c leaves a correlated residual	Yes — balance panel within regime (drops ∼40% of single-cell obs)
C6	No within-cell regressor heterogeneity	Firm-level OLS uses within-cell variation (W_{XX}, W_{XY}) that cell-level OLS cannot see	**No** — inherent to real data